

Understanding Generative Adversarial Networks from a mathematical view

Pipi Hu

Beijing Institute of Mathematical Sciences and Applications

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What I cannot create, I do not understand.

Outline

- 1 Preliminary: Generative Modeling and Adversarial Training
- 2 Theoretical Foundations: Minimax Game and Nash Equilibrium
- 3 Loss Functions and Training Objectives
- 4 WGAN
- 5 GAN Architectures and Design Principles
- 6 Training Challenges and Solutions
- 7 Evaluation Metrics and Methods
- 8 The GAN is dead; long live the GAN! A Modern Baseline GAN (R3GAN)
- 9 Open Questions and Future Research

The Generative Modeling Problem

Given a dataset $\{x_i\}_{i=1}^n$ sampled from an unknown distribution $p_{data}(x)$, how to learn a model that can generate new samples from $p_{data}(x)$?

- **Explicit modeling:** Learn $p_{\theta}(x)$ directly
 - Maximum Likelihood Estimation (MLE)
 - Variational Autoencoders (VAEs)
 - Normalizing Flows
- **Implicit modeling:** Learn a generator $G_{\theta}(z)$ that maps noise to data
 - Generative Adversarial Networks (GANs)
 - Diffusion Models

The Adversarial Training Paradigm

The key insight of GANs: **Two-player minimax game**

- **Generator** $G_\theta(z)$: Maps noise $z \sim p_z$ to data space
- **Discriminator** $D_\phi(x)$: Distinguishes real from fake data
- **Objective**: Generator tries to fool discriminator, discriminator tries to detect fakes

The minimax objective:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))] \quad (1)$$

Why Adversarial Training?

Traditional generative models face challenges:

- **Explicit likelihood:** Hard to compute for complex distributions
- **Mode collapse:** Tend to generate similar samples
- **Blurry samples:** MLE often produces average-looking samples

GANs offer advantages:

- **No explicit likelihood:** Avoid normalization issues
- **Sharp samples:** Adversarial training encourages realistic details
- **Flexible architecture:** Can use any differentiable network

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The Minimax Game Formulation

For fixed generator G , the optimal discriminator is:

$$D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_g(x)} \quad (2)$$

Proof.

The discriminator loss for fixed G :

$$V(D, G) = \int_x p_{data}(x) \log D(x) + p_g(x) \log(1 - D(x)) dx \quad (3)$$

Taking derivative w.r.t. $D(x)$ and setting to zero:

$$\frac{p_{data}(x)}{D(x)} - \frac{p_g(x)}{1 - D(x)} = 0 \Rightarrow D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_g(x)} \quad (4)$$



Global Optimum Analysis

When $p_g = p_{data}$, the optimal discriminator becomes:

$$D_G^*(x) = \frac{1}{2} \quad (5)$$

The global minimum of the generator loss is:

$$C(G) = \max_D V(D, G) = -\log(4) + 2 \cdot JSD(p_{data} \| p_g) \quad (6)$$

where JSD is the Jensen-Shannon divergence.

Proof.

Substituting D_G^* into $V(D, G)$:

$$\begin{aligned} C(G) &= \mathbb{E}_{x \sim p_{data}} \left[\log \frac{p_{data}(x)}{p_{data}(x) + p_g(x)} \right] + \mathbb{E}_{x \sim p_g} \left[\log \frac{p_g(x)}{p_{data}(x) + p_g(x)} \right] \\ &= -\log(4) + KL(p_{data} \| \frac{p_{data} + p_g}{2}) + KL(p_g \| \frac{p_{data} + p_g}{2}) \\ &= -\log(4) + 2 \cdot JSD(p_{data} \| p_g) \end{aligned} \quad (7)$$

Convergence Analysis

Theorem: If G and D have enough capacity, and at each step the discriminator is allowed to reach its optimum given G , then p_g converges to p_{data} .

- **Sufficient capacity:** Networks can represent the optimal functions
- **Perfect discriminator:** D reaches optimum at each step
- **Generator updates:** G is updated to minimize $C(G)$

However, in practice:

- Networks have limited capacity
- Discriminator is not trained to optimality
- Training dynamics can be unstable

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Original GAN Loss

The original GAN uses the binary cross-entropy loss:

$$\mathcal{L}_{GAN} = \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))] \quad (8)$$

Problems with original loss:

- **Gradient vanishing:** When D is too good, gradients vanish
- **Mode collapse:** Generator finds one mode and sticks to it
- **Training instability:** Oscillations between G and D

Non-saturating loss (alternative for generator):

$$\mathcal{L}_G^{NS} = -\mathbb{E}_{z \sim p_z} [\log D(G(z))] \quad (9)$$

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WGAN: What goes wrong with vanilla GANs?

- Objective uses **Jensen–Shannon (JS)** divergence between p_θ and p_D .
- When supports are disjoint or lie on low-dimensional manifolds, JS is **maxed out** and gradients vanish.
- Symptoms: **mode collapse**, **training instability**, sensitivity to architecture and hyperparameters.

WGAN: Wasserstein-1 (Earth-Mover) distance

$$\mathcal{W}(p_\theta, p_{\mathcal{D}}) = \inf_{\gamma \in \Pi(p_\theta, p_{\mathcal{D}})} \mathbb{E}_{(\mathbf{x}, \tilde{\mathbf{x}}) \sim \gamma} [\|\mathbf{x} - \tilde{\mathbf{x}}\|], \quad (10)$$

where Π are couplings with marginals p_θ and $p_{\mathcal{D}}$. **Why it helps:**

- Provides *meaningful* distances even when supports do not overlap.
- Changes *continuously* with generator parameters θ .
- Leads to **usable gradients** almost everywhere.

WGAN: Kantorovich–Rubinstein duality

$$\mathcal{W}(p_\theta, p_{\mathcal{D}}) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\mathbb{E}_{\mathbf{x} \sim p_{\mathcal{D}}} [f(\mathbf{x})] - \mathbb{E}_{\mathbf{x} \sim p_\theta} [f(\mathbf{x})] \right). \quad (11)$$

WGAN idea

Learn a **1-Lipschitz critic** $\mathcal{D}$ to approximate the supremum and minimize $\mathcal{W}$ by moving p_θ .

WGAN: Optimization problem

Critic (*not a classifier*):

$$\max_{\psi : \|\mathcal{D}\|_{\text{Lip}} \leq 1} \mathbb{E}_{\mathbf{x} \sim p_{\mathcal{D}}}[\mathcal{D}(\mathbf{x})] - \mathbb{E}_{\mathbf{z} \sim p(\zeta)}[\mathcal{D}(\mathcal{G}(\mathbf{z}))]. \quad (12)$$

Generator:

$$\min_{\theta} \mathbb{E}_{\mathbf{z} \sim p(\zeta)}[\mathcal{D}(\mathcal{G}(\mathbf{z}))]. \quad (13)$$

- The critic's output is a **score** (real number), not a probability.
- Enforcing 1-Lipschitzness is *crucial* (see next slide).

WGAN: How to enforce 1-Lipschitz?

Weight clipping (original WGAN)

- Constrain each weight to $[-c, c]$.
- Simple but can underfit or cause gradient issues.

Gradient penalty (WGAN-GP)

$$\lambda \mathbb{E}_{\hat{\mathbf{x}}=t\mathbf{x}+(1-t)\tilde{\mathbf{x}}} \left(\|\nabla_{\hat{\mathbf{x}}} \mathcal{D}(\hat{\mathbf{x}})\|_2 - 1 \right)^2. \quad (14)$$

- Encourages $\|\nabla \mathcal{D}\| \approx 1$ on lines between real and fake.
- Typically far more stable than clipping.

Least Squares GAN (LSGAN)

LSGAN uses the least squares loss:

$$\min_D \mathcal{L}_D^{LS} = \frac{1}{2} \mathbb{E}_{x \sim p_{data}} [(D(x) - 1)^2] + \frac{1}{2} \mathbb{E}_{z \sim p_z} [(D(G(z)))^2] \quad (15)$$

$$\min_G \mathcal{L}_G^{LS} = \frac{1}{2} \mathbb{E}_{z \sim p_z} [(D(G(z)) - 1)^2] \quad (16)$$

Advantages:

- **Stable gradients:** No vanishing gradient problem
- **Better quality:** Produces higher quality samples
- **Faster convergence:** More stable training dynamics

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DCGAN: Deep Convolutional GAN

Key architectural guidelines:

- **Replace pooling** with strided convolutions
- **Remove fully connected layers** (except first/last)
- **Batch normalization** in both G and D
- **ReLU** in generator, **LeakyReLU** in discriminator
- **Remove bias** from batch norm layers

Generator architecture:

- Input: 100-dim noise z
- Project to $4 \times 4 \times 1024$ feature maps
- Upsample through transposed convolutions
- Output: $64 \times 64 \times 3$ RGB image

Progressive Growing of GANs

Train GANs progressively from low to high resolution:

- Start with 4×4 images
- Gradually add layers for higher resolution
- Smooth transition between resolutions

Benefits:

- **Stable training**: Easier to train at low resolution
- **High quality**: Can generate very high resolution images
- **Faster convergence**: Progressive training is more efficient

Key techniques:

- Fade-in new layers gradually
- Use minibatch standard deviation
- Equalized learning rate

StyleGAN: Style-Based Generator

StyleGAN introduces style-based architecture:

- **Mapping network:** Transforms z to intermediate latent w
- **Style modulation:** Each layer can be modulated by style
- **Noise injection:** Adds stochastic variation
- **Progressive growing:** Builds up resolution gradually

Key innovations:

$$y = (w \cdot s) \cdot x + b \quad (17)$$

where w is the style vector, s is the style scale, x is the feature map.

Benefits:

- **Better control:** Separate content and style
- **Higher quality:** More realistic generation
- **Interpolation:** Smooth latent space

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Common Training Problems

Mode Collapse:

- Generator produces limited variety of samples
- Discriminator becomes too strong
- Generator finds one "safe" mode

Training Instability:

- Losses oscillate wildly
- Generator and discriminator don't converge
- Sensitive to hyperparameters

Vanishing Gradients:

- Discriminator becomes too good
- Generator receives no useful gradients
- Training stagnates

Solutions to Training Problems

For Mode Collapse:

- **Unrolled GANs**: Look ahead in discriminator updates
- **Minibatch discrimination**: Encourage diversity
- **Feature matching**: Match feature statistics

For Training Instability:

- **Two time-scale update rule**: Different learning rates
- **Spectral normalization**: Constrain discriminator
- **Gradient penalty**: Enforce Lipschitz constraint

For Vanishing Gradients:

- **Wasserstein loss**: More stable gradients
- **Least squares loss**: Better gradient properties
- **Hinge loss**: Alternative to cross-entropy

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Inception Score (IS):

$$IS = \exp(\mathbb{E}_{x \sim p_g}[KL(p(y|x) \| p(y))]) \quad (18)$$

where $p(y|x)$ is the class probability from Inception network.

Fréchet Inception Distance (FID):

$$FID = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) \quad (19)$$

where (μ_r, Σ_r) and (μ_g, Σ_g) are mean and covariance of real and generated features.

Precision and Recall:

- **Precision:** Quality of generated samples
- **Recall:** Coverage of real data distribution

Qualitative Evaluation Methods

Interpolation:

- Linear interpolation in latent space
- Should produce smooth transitions
- Tests continuity of learned manifold

Arithmetic in Latent Space:

- $G(z_1) + G(z_2) - G(z_3) \approx G(z_1 + z_2 - z_3)$
- Tests linear structure of latent space

Truncation Trick:

- Vary the magnitude of latent vectors
- Trade-off between quality and diversity
- Useful for controlling generation

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R3GAN in one picture: fix the landscape & the dynamics

Goal. Make GAN a *convergent* game with strong fidelity at 1-NFE.

Two steps:

- 1 **Relativistic pairing (RpGAN)** improves the *loss landscape* by comparing real vs. fake relatively:

$$L_{\text{Rp}}(\theta, \psi) = \mathbb{E}_{x \sim p_{\mathcal{D}}, z \sim p(\zeta)} f(D_{\psi}(G_{\theta}(z)) - D_{\psi}(x)),$$

which anchors decision boundaries around each real sample and mitigates mode dropping.

- 2 **Zero-centered gradient penalties (R1+R2)** fix the *training dynamics*:

$$R1(\psi) = \frac{\gamma}{2} \mathbb{E}_{x \sim p_{\mathcal{D}}} [\|\nabla_x D_{\psi}(x)\|_2^2], \quad R2(\theta, \psi) = \frac{\gamma}{2} \mathbb{E}_{x \sim p_{\theta}} [\|\nabla_x D_{\psi}(x)\|_2^2].$$

Why RpGAN helps: relative scoring removes bad minima

Classic f-GAN:

$$L(\theta, \psi) = \mathbb{E}_z[f(D_\psi(G_\theta(z)))] + \mathbb{E}_x[f(-D_\psi(x))].$$

Mode dropping corresponds to spurious local minima/flat regions.

RpGAN:

$$L_{\text{Rp}}(\theta, \psi) = \mathbb{E}_{x,z} f(D_\psi(G_\theta(z)) - D_\psi(x)).$$

- Each real x contributes a local margin against nearby fakes $\Rightarrow$ fewer mode-drop basins.
- *However*, unregularized RpGAN can still diverge under gradient descent (unstable dynamics).

R1+R2: enforcing the correct fixed point (zero slope at equilibrium)

At equilibrium $p_\theta = p_{\mathcal{D}}$ we want the critic's spatial gradient to *vanish*:

$$\nabla_x D_\psi(x) = 0 \quad \text{for } x \in \text{supp}(p_\theta) = \text{supp}(p_{\mathcal{D}}).$$

Zero-centered penalties achieve exactly this:

$$R1(\psi) = \frac{\gamma}{2} \mathbb{E}_{x \sim p_{\mathcal{D}}} [\|\nabla_x D_\psi(x)\|^2], \quad R2(\theta, \psi) = \frac{\gamma}{2} \mathbb{E}_{x \sim p_\theta} [\|\nabla_x D_\psi(x)\|^2].$$

Contrast with WGAN-GP (“1-GP”): encourages $\|\nabla_x D\| \approx 1$ along real-fake lines, which *does not* enforce the zero-slope fixed point and can sustain rotational dynamics near the optimum.

Local convergence via Jacobian of the game dynamics (informal)

Define the training field

$$v(\theta, \psi) = \begin{pmatrix} -\nabla_{\theta} L_{\text{Rp}}(\theta, \psi) \\ \nabla_{\psi} L_{\text{Rp}}(\theta, \psi) + \nabla_{\psi} (\mathbf{R1} + \mathbf{R2}) \end{pmatrix}, \quad J = \nabla v(\theta^*, \psi^*).$$

First-order update $(\theta, \psi) \leftarrow (\theta, \psi) + h v(\theta, \psi)$.

Claim (informal): With RpGAN + **R1+R2**, J has eigenvalues with negative real parts near (θ^*, ψ^*) for small h :

$$\Re \sigma(J) < 0 \implies \text{linearized stability (no cycling)}.$$

Intuition: 0-GP damps the critic blocks (adds positive-definite terms), kills the rotational component that causes oscillation, and aligns the fixed point with $\nabla_x D = 0$.

Putting it together: the R3GAN objective & loop

Critic loss (one minibatch):

$$\mathcal{L}_{\text{critic}} = -\frac{1}{m} \sum_{i=1}^m f(D_{\psi}(\tilde{x}_i) - D_{\psi}(x_i)) + \frac{\gamma}{2} (\|\nabla D_{\psi}(x_i)\|^2 + \|\nabla D_{\psi}(\tilde{x}_i)\|^2),$$

where $\tilde{x}_i = G_{\theta}(z_i)$.

Generator loss:

$$\mathcal{L}_{\text{gen}} = -\frac{1}{m} \sum_{i=1}^m f(D_{\psi}(G_{\theta}(z_i)) - D_{\psi}(x_i)).$$

Recipe: several critic steps per gen step; Adam with $\beta_1=0$; moderate γ ; balanced batch sizes.

Why the engineering works (enabled by the math)

Once the *game* is stable, R3GAN drops many ad-hoc tricks and uses a clean backbone:

- **No normalization**; fix-up initialization to keep variance controlled.
- **Bilinear** up/down-sampling; avoid transposed-conv artifacts.
- Per-resolution **Transition** (resample+1x1) + two **1–3–1 residual** blocks.
- **Grouped conv** + **inverted bottlenecks** for efficient width.
- Symmetric G/D; LeakyReLU; small LR for critic.

Outcome: strong 1-NFE quality (FFHQ/CIFAR/ImageNet) and full mode coverage on StackedMNIST *without* StyleGAN-style tricks.

- ① **Landscape:** RpGAN's relative scoring reduces mode-drop minima:

$$L_{\text{Rp}}(\theta, \psi) = \mathbb{E}_{x,z} f(D(G(z)) - D(x)).$$

- ② **Dynamics:** Zero-centered penalties make the fixed point correct ($\nabla_x D = 0$) and the Jacobian stable:

$$R1 + R2 \Rightarrow \Re \sigma(J) < 0.$$

- ③ **Engineering:** with a convergent loss, a minimalist modern backbone achieves SOTA-level 1-NFE results.

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Open Questions

If you are interested in these questions, please feel free to write an email to me (hpp@bimsa.cn) with your comments.

1. **Theoretical understanding:** Why do GANs work despite the minimax game being non-convex?
2. **Mode collapse:** Can we guarantee that GANs will cover all modes of the data distribution?
3. **Evaluation metrics:** How can we better evaluate the quality and diversity of generated samples?
4. **Training stability:** What are the fundamental principles for stable GAN training?
5. **Generalization:** How do GANs generalize beyond the training distribution?

Welcome inputs

Any comments?

Welcome your inputs!